

Multi-Tenant RAG Architecture

AI ENGINEER CANDIDATE ASSESSMENT

Project Overview

Welcome to the practical portion of the AI Engineering interview process. We are evaluating your ability to design and build a multi-tenant Generative AI backend.

You have been provided with two PDF handbooks belonging to two fictional clients: **Acme Corp** and **Globex Corporation**. Your task is to build a RAG (Retrieval-Augmented Generation) system that accurately isolates and serves data based on the selected company.

Objective

Demonstrate your ability to build a data ingestion pipeline and a multi-tenant API capable of accurately routing user queries to the correct document corpus.

Core Requirements

1 Data Ingestion

Build a pipeline to parse, chunk, and embed both provided PDFs into a vector database. You must architect the ingestion process and database schema to actively support multiple distinct tenants.

2 Multi-Tenant Query System

Build a system where a user can select a specific company and ask a question. Expose an API endpoint that accepts this query alongside the company identifier (tenant selection).

The system must retrieve context and generate an answer *strictly* based on the selected company's data. Cross-company data contamination must be avoided.

Deliverables

Please submit a link to a public GitHub repository containing your complete solution. Your submission must include the following:

- **The Source Code:** All necessary code for both the ingestion pipeline and the API backend.
- **Code Documentation:** Clean, well-structured, and documented code explaining your logic and implementation details.
- **README.md:** A comprehensive markdown file at the root of your repository. This file must include clear instructions on how to install dependencies, run the ingestion script, and start the system locally. Include example API requests or usage instructions.

Good luck! We look forward to reviewing your approach to multi-tenant Generative AI architecture.